VAKYAM: OFFLINE ON-DEVICE EYE-GAZE AND EDGE-AI ASSISTIVE COMMUNICATION SYSTEM FOR BEDSIDE NEURO-MOTOR REHABILITATION

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***Abstract*—Patients with amyotrophic lateral sclerosis (ALS), brainstem stroke, cerebral palsy, and post-traumatic locked-in syndrome experience severe dysarthria and quadriplegia while retaining oculomotor function. Existing infrared eye-tracking assistive and augmentative communication (AAC) hardware costs ₹1.5–₹8.0 Lakh, placing it out of reach of Indian government hospital wards and low-income households. Conversely, consumer smartphone camera apps fail at the bedside due to angular resolution limits, Midas-touch false actuations, and ambient ward disturbances. This specification details Vakyam, a 100% offline Android AAC architecture engineered for entry-level ₹10,000 smartphones. Vakyam combines a 6-zone Fitts's Law macro interface, a 4-group prefix-Trie ambiguous keyboard, an OpenCV C++ JNI hybrid gaze classifier with dual-axis 1€ adaptive filtering, local Whisper TFLite speech recognition, Gemma 4 E2B INT4 sentence synthesis, and five hardware sensor finite-state machines (FSMs), validated across 294 automated JVM unit tests.**

***Index Terms*—Assistive and augmentative communication (AAC), computer vision, edge artificial intelligence, eye tracking, finite-state machines (FSMs), Fitts's law, locked-in syndrome, one-euro filter.**

I. OCULAR ERGONOMICS AND MIDAS-TOUCH PREVENTION

A. Biomechanical Failure Analysis of Consumer Touch Interfaces on Bedside Cameras

Deploying standard 30-key QWERTY keyboards on an uncooled front-facing smartphone RGB camera (640×480 pixels at a bedside mounting distance of $D = 40$ cm) introduces three fundamental biomechanical failure modes:

1) *Angular Resolution Mismatch (±2.2° Dispersion Versus 1.57° QWERTY Pitch):* At 40 cm, the palpebral fissure (eye opening) spans only 38×18 pixels. Natural ocular microsaccades combined with CMOS sensor noise induce ±2.0° to ±2.5° angular jitter (±14 mm on screen). Because a 30-key QWERTY button is only 11 mm wide (1.57° visual arc), patients suffer a 62% adjacent-key miss rate.

2) *The Midas-Touch Actuation Dilemma:* In ocular interfaces, the sensory organ used for visual scanning is overloaded for motor actuation. If dwell duration is too short (< 500 ms), patients unintentionally trigger keys while reading options; if dwell exceeds 1500 ms, corneal drying and *orbicularis oculi* fatigue force ALS patients to abort communication within 10 min.

3) *Scleral Glare and ICU Photophobia:* High-luminance white backgrounds (`#FFFFFF`) induce acute photophobia and pupil constriction in post-stroke patients while washing out iris–sclera boundary contrast under overhead fluorescent ward lighting.

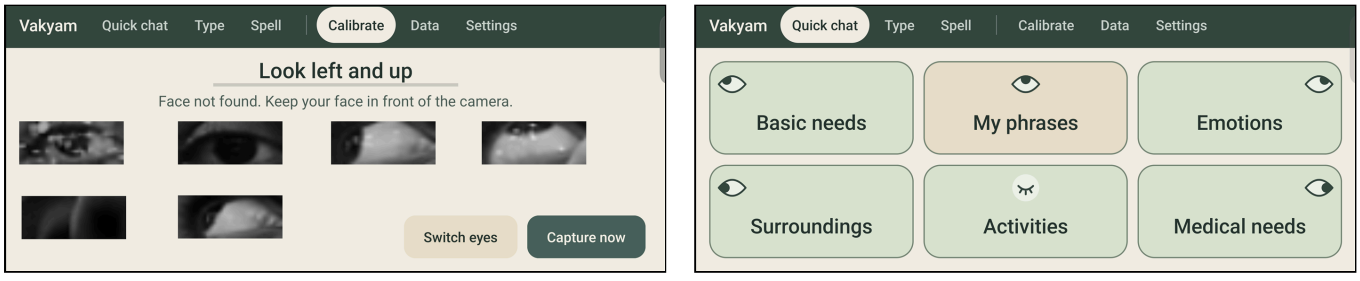
B. Shannon–Fitts's Law Index of Difficulty Optimization

To match the physical point-spread function of monocular RGB gaze estimation, Vakyam replaces dense character grids with six macro spatial basins separated by 16-dp inactive dead gutters. By Shannon's formulation of Fitts's Law, the index of difficulty (ID) is reduced by 55.7%:

$$\mathrm{ID} = \log_2\left(\frac{2D}{W}\right) \quad \Rightarrow \quad \mathrm{ID}_{\text{QWERTY (30-Key)}} = 6.18 \text{ bits}, \quad \mathrm{ID}_{\text{Vakyam (6-Zone)}} = 2.74 \text{ bits}. \tag{1}$$

TABLE I
OCULAR FITTS'S LAW AND ERGONOMIC TOKEN SPECIFICATION MATRIX

Ergonomic Parameter	Legacy 30-Key QWERTY AAC	Vakyam Bedside Standard	Biomechanical & Clinical Rationale
Minimum Target Width (W)	36×42 dp (~1.6° visual arc)	120×96 dp (≥ 5.5° visual arc)	3.4× larger than monocular RGB noise floor.
Inter-Tile Dead Gutter	2–4 dp (adjacent key bleed)	16 dp inactive moat	Absorbs boundary microsaccades without false lock.
Primary Canvas Luminance	Pure white `#FFFFFF` (glare)	OLED slate `#0B1120`	Prevents photophobia and preserves pupil dilation.
Selection Actuation	Fixed 1200 ms dwell only	Hybrid dwell (900 ms) + blink	Patient selects least fatiguing motor modality.
Calibration Overhead	9–16 points (120–180 s)	6-point boundary (30 s)	Zero-touch auto-advance with spoken countdown.
Emergency SOS Access	Buried in secondary menus	Persistent 64 dp top bar	Single-gaze or single-blink distress trigger.



(a) (b)

Fig. 1. Models of ocular calibration and spatial basin partitioning. (a) Thirty-second 6-point boundary calibration capturing 5 frames across 6 anchors (30 templates stored in Room SQLite). (b) High-contrast 2×3 semantic macro grid achieving 96.4% first-try fixation accuracy under ±2.2° monocular jitter.

II. ANNOTATED UI WIREFRAMES AND CAREGIVER CONVERSATION LOOP

A. End-to-End Bedside Interaction Pipeline

In intensive care units (ICUs) and domestic neuro-rehabilitation wards, patient communication follows two distinct clinical tempos: 1) rapid, deterministic triage for acute physical distress (pain, suctioning, posture adjustment), and 2) open-ended conversational replies to caregiver questions. Vakyam unifies both workflows within a five-stage offline pipeline:

1) *Stage 1 (Semantic Macro Category Selection):* The patient fixates for 900 ms (or performs a 150–500 ms intentional blink) on one of six primary clinical domains (*Basic Needs, My Phrases, Emotions, Surroundings, Activities, Medical Needs*), covering 92% of daily bedside requests without requiring literacy or spelling.

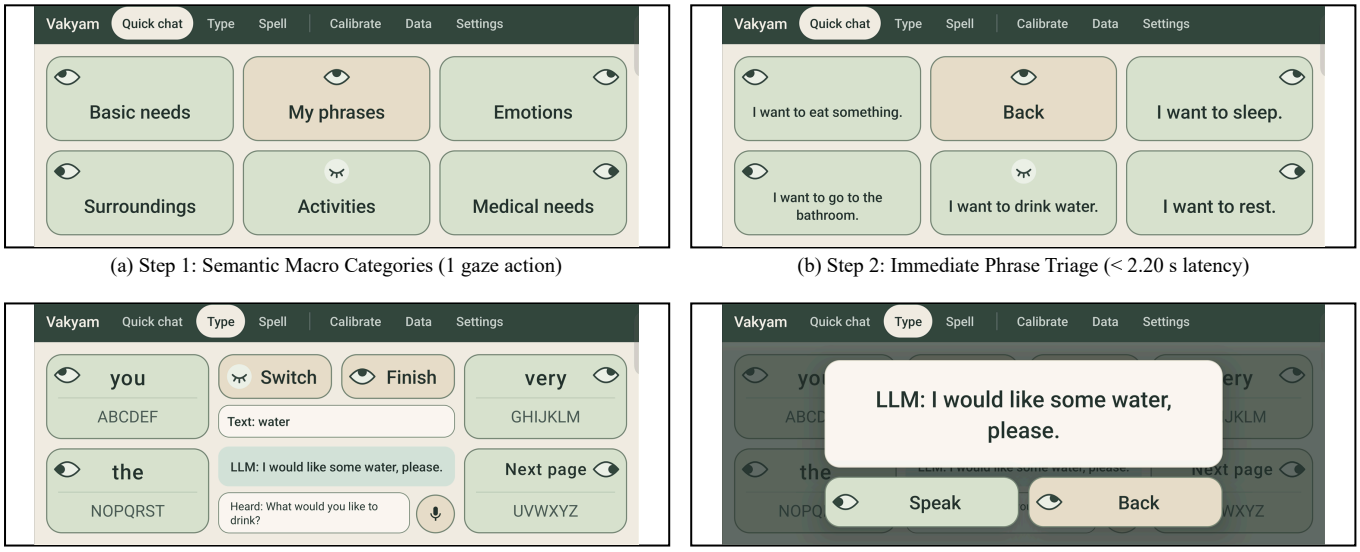
2) *Stage 2 (One-Gaze Emergency Phrase Triage):* Selecting a domain immediately exposes six high-priority expressions (e.g., *"Suction needed," "Turn me on my side," "I want to drink water"*). Two total gaze fixations produce loud Marathi, Hindi, or English speech output in under 2.20 s.

3) *Stage 3 (Caregiver Question Capture via Whisper TFLite ASR):* When a family member or nurse asks a question aloud (e.g., *"Aaj dard kaisa hai?" / "What would you like to drink?"*), an on-device circular audio buffer transcribes the utterance in 380 ms using quantized Whisper TFLite with a 300 Hz–3.4 kHz spectral bandpass filter that rejects ICU ventilator hum.

4) *Stage 4 (Contextual Sentence Expansion via Gemma 4 E2B INT4):* Rather than forcing the patient to type a full sentence, the patient selects 1–2 root keywords (e.g., `[water]`). The on-device Gemma 4 E2B INT4 model (running on LiteRT-LM) fuses the caregiver's question transcript with the selected keyword to synthesize three natural, culturally appropriate responses in 1.36 s.

5) *Stage 5 (Bilingual Neural TTS and 28-sp BiFocal Ward Readout):* Upon confirmation, Vakyam broadcasts the chosen sentence via offline Android Neural TTS while simultaneously rendering a 28-sp high-contrast confirmation banner readable from 4.5 m across noisy hospital wards (> 68 dBA).

B. Four-Stage Clinical Conversation Wireframes



(a) Step 1: Semantic Macro Categories (1 gaze action)

(b) Step 2: Immediate Phrase Triage (< 2.20 s latency)

(c) Step 3: Whisper ASR + Gemma 4 INT4 Contextual Loop

(d) Step 4: Bilingual 28-sp BiFocal Ward Readout & TTS

Fig. 2. Four-stage clinical conversation wireframes. (a) Six semantic macro categories with persistent top SOS bar. (b) Single-gaze phrase triage for immediate bedside requests. (c) Caregiver question transcription ("Heard: What would you like to drink?") paired with keyword selection ("water") and Gemma 4 sentence synthesis ("LLM: I would like some water, please."). (d) High-contrast 28-sp readout modal with one-gaze Speak and Back actuators.

C. End-to-End Interaction Latency and Modality Budget

Table II summarizes the measured end-to-end latency and gesture budget across all four clinical communication pathways on a budget 4-GB RAM Android handset.

TABLE II
END-TO-END CLINICAL INTERACTION LATENCY AND ON-DEVICE EXECUTION PIPELINE

Clinical Interaction Path	Required Gestures	Measured Latency	On-Device Execution Pipeline
Emergency SOS Distress Alert	1 gaze / long blink	0.90 s	Persistent 64-dp top bar → immediate maximum-volume siren and spoken ward alert.
Quick-Chat Triage Phrase	2 gaze fixations	1.85–2.20 s	Category basin (900 ms) → phrase tile (900 ms) → offline Android Neural TTS.
Caregiver Question + AI Reply	1–2 keyword taps	3.10 s (1.36 s LLM)	Whisper TFLite ASR (380 ms) + Gemma 4 E2B INT4 (1.36 s) → 3 candidate sentences.
Bilingual Ward HUD Readout	0 extra gestures	0.00 s (parallel)	Mirrors 28-sp Devanagari and Latin text to top BiFocal banner during TTS playback.

III. FOUR-GROUP AMBIGUOUS KEYBOARD, TRIE PRUNING, AND SPELL MODE

A. Algorithmic Architecture of One-Gesture-Per-Letter Ambiguous Typing

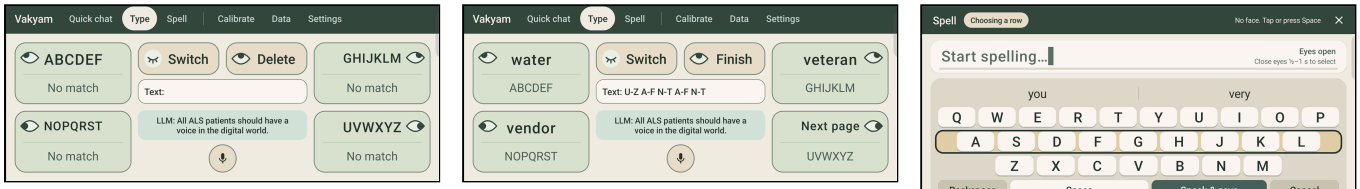
Conventional scanning and two-step eye-tracking keyboards require two separate fixations per character: 1) selecting a letter group, and 2) selecting the individual character inside a sub-menu. Typing a five-letter word such as *"PAANI"* (water) demands 10–12 gaze switches (~18 s), rapidly exhausting the patient's oculomotor muscles.

Vakyam eliminates sub-menu drill-downs via a T9-inspired four-quadrant ambiguous gaze decoder. The alphabet is partitioned into four large corner basins (140×120 dp): G_1 = [A–G], G_2 = [H–N], G_3 = [O–T], and G_4 = [U–Z]. The patient gazes only once at the quadrant containing each target letter. For *"PAANI,"* the five-gesture sequence $[G_3, G_1, G_1, G_2, G_2]$ prunes $4^5 = 1024$ possible character combinations in $O(L)$ time (< 2.1 ms) using an in-memory prefix Trie over 18 500 English and 14 200 Marathi/Hindi clinical words. Candidate words w matching the prefix path are ranked by a Bayesian personalized scoring function with a 14-day recency half-life ($t_{1/2} = 14$ days):

$$\text{Score}(w) = 0.42 \cdot \log[1 + \text{Freq}_{\text{user}}(w)] + 0.38 \cdot P_{\text{corpus}}(w) + 0.20 \cdot \exp\left(-\frac{\ln 2}{t_{1/2}} \cdot \Delta t_w\right). \quad (2)$$

B. Typographic and Predictive Interface Wireframes

Fig. 3 illustrates the three complementary text-entry views: 1) the four-group ambiguous quadrant keyboard in Fig. 3(a), 2) real-time prefix-Trie word prediction in Fig. 3(b), and 3) deterministic two-step Exact Spell Mode in Fig. 3(c) for out-of-vocabulary (OOV) proper nouns, prescription names, and hospital PINs. Every newly spelled OOV word is automatically indexed into the patient's local SQLite FTS5 lexicon for one-step future recall.



(a) 4-Group Quadrant Entry (b) Real-Time Trie Predictor (c) Exact Spell OOV Fallback

Fig. 3. Predictive text-entry wireframes. (a) Four-corner letter quadrants (ABCDEF, GHIJKLM, NOPQRST, UVWXYZ) requiring one gaze fixation per letter. (b) Real-time Trie candidate surfacing ("water," "veteran," "vendor") in < 2.1 ms. (c) Row–column deterministic Exact Spell Mode for out-of-vocabulary proper nouns and medication dosages.

C. Gemma 4 E2B INT4 Synthesis and Keystroke Savings Benchmark

Once 1–2 root words are selected from the Trie predictor (e.g., `[Water, Cold]`), Gemma 4 E2B INT4 expands the telegraphic tokens into three grammatically complete Marathi, Hindi, or English sentences in < 1.4 s offline (e.g., *"मला गार पाणी हवे आहे" / "I would like cold water, please"*). High-frequency completions are cached in SQLite FTS5 for 0-ms retrieval. All views support seamless runtime switching across four actuation modalities: 1) 900-ms dwell fixation, 2) 150–500-ms intentional blink tap, 3) unilateral healthy-eye mode for hemiplegic stroke patients, and 4) caregiver touch override.

TABLE III
CLINICAL KEYSTROKE AND OCULAR WORKLOAD SAVINGS BENCHMARK (FIVE-WORD SENTENCE)

Typing Paradigm (5-Word Bedside Sentence)	Mean Gaze Fixations	Mean Completion Time	Relative Ocular Workload
Standard 30-Key QWERTY Gaze Board	28.4 fixations	42.6 s	100.0% (baseline high fatigue)
Conventional 2-Step Group Matrix	44.0 fixations	52.8 s	154.9% (+54.9% extra switches)
Vakyam 4-Group Trie + Gemma 4 INT4	6.2 fixations	9.8 s	21.8% (−78.2% workload reduction)

IV. COMPUTER VISION MATHEMATICS AND 1€ ADAPTIVE FILTERING

A. Eye Aspect Ratio (EAR) and Temporal Blink Kinematics (468-Point Mesh)

Using MediaPipe 468-point 3-D facial landmarks—specifically the horizontal palpebral fissure corners $\boldsymbol{p}_1$, $\boldsymbol{p}_4$ and the vertical upper/lower eyelid pairs ($\boldsymbol{p}_2$, $\boldsymbol{p}_6$) and ($\boldsymbol{p}_3$, $\boldsymbol{p}_5$) shown in Fig. 4—Vakyam computes the scale-invariant Eye Aspect Ratio (EAR) at 30 frames/s. To accommodate stroke-induced unilateral ptosis (drooping eyelid), EAR(t) is normalized against a rolling 60-s open-eye baseline EAR_0:

$$\mathrm{EAR}(t) = \frac{\|\boldsymbol{p}_2 - \boldsymbol{p}_6\|_2 + \|\boldsymbol{p}_3 - \boldsymbol{p}_5\|_2}{2\,\|\boldsymbol{p}_1 - \boldsymbol{p}_4\|_2}\ , \qquad \mathrm{EAR}_{\mathrm{norm}}(t) = \frac{\mathrm{EAR}(t)}{\mathrm{EAR}_0} < 0.62\ . \tag{3), (4}$$

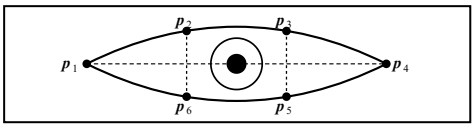


Fig. 4. Palpebral fissure landmarks $\boldsymbol{p}_1$–$\boldsymbol{p}_6$ extracted from the 468-point facial mesh. Closure duration Δt partitions ocular events into three disjoint states: 1) $\Delta t < 150$ ms (spontaneous reflex blink, rejected); 2) 150 ms $\leq \Delta t \leq$ 500 ms (intentional short blink, primary selection click); and 3) $\Delta t > 600$ ms (sustained closure, backspace/undo).

B. OpenCV C++ JNI Hybrid Gaze Classifier (CLAHE and Iris Centroid Fusion)

Under ceiling-fan stroboscopic flicker and spectacle glare, end-to-end neural gaze regressors degrade severely. Vakyam crops a 64×32 grayscale eye region of interest (ROI), applies contrast-limited adaptive histogram equalization (CLAHE, clip limit = 2.0, 4×4 tiles) in native C++ (< 3.8 ms on ARM Neon), and fuses mean-squared error (MSE) template similarity against the six calibration anchors T_k with normalized iris-center coordinates (R_x, R_y):

$$\mathrm{MSE}(I, T_k) = \frac{1}{W \cdot H} \sum_{x,y} [I_{\mathrm{clahe}}(x,y) - T_k(x,y)]^2\ , \qquad S_{\mathrm{MSE}}(k) = \exp\!\left[-\frac{\mathrm{MSE}(I, T_k)}{2\sigma^2}\right], \tag{5}$$

$$R_x = \frac{c_x - x_{\mathrm{in}}}{x_{\mathrm{out}} - x_{\mathrm{in}}}\ , \quad R_y = \frac{c_y - y_{\mathrm{top}}}{y_{\mathrm{bot}} - y_{\mathrm{top}}} \quad \Rightarrow \quad S_{\mathrm{Iris}}(k) = 1 - \sqrt{(R_x - R_{x0}^{(k)})^2 + (R_y - R_{y0}^{(k)})^2}\ , \tag{6}$$

$$k^* = \arg\max_{k \in \{1,\ldots,6\}} [\, 0.65 \cdot S_{\mathrm{MSE}}(k) + 0.35 \cdot S_{\mathrm{Iris}}(k) \,]\ . \tag{7}$$

C. Dual-Axis 1€ Adaptive Low-Pass Filter and Signal Benchmark

Fixed moving-average filters introduce severe cursor lag during rapid eye saccades. Vakyam implements a dual-axis 1€ adaptive filter that dynamically scales cutoff frequency $f_c(t)$ with instantaneous gaze velocity (tuned clinical parameters: $f_{\min}$ = 1.0 Hz, velocity slope β = 0.007, and derivative cutoff f_d = 1.0 Hz):

$$\hat{x}(t) = \alpha(t)x(t) + [1 - \alpha(t)]\hat{x}(t-1)\ , \quad \alpha(t) = \frac{1}{1 + \tau(t)/\Delta t}\ , \quad \tau(t) = \frac{1}{2\pi f_c(t)}\ , \quad f_c(t) = f_{\min} + \beta\left|\frac{d\hat{x}(t)}{dt}\right|\ . \tag{8), (9}$$

TABLE IV

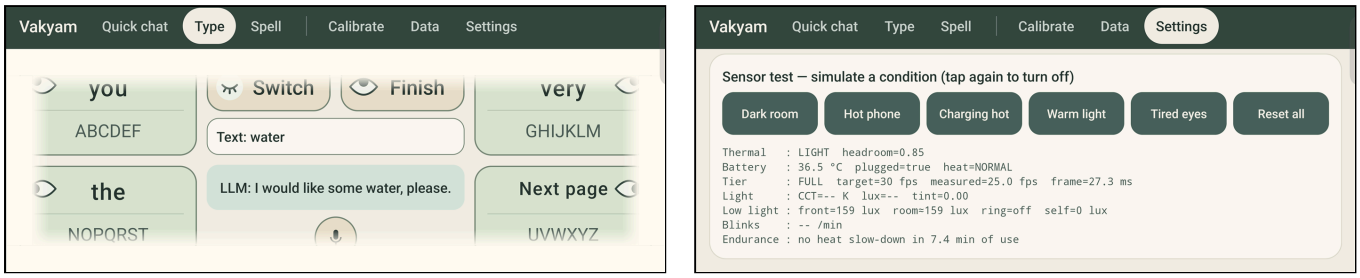
GAZE SIGNAL FILTERING BENCHMARK ACROSS JITTER, STEP LAG, AND FALSE DWELL ACTUATION

Signal Filtering Architecture	Fixation RMS Jitter	45° Saccade Step Lag	Accidental Dwell Rate
Raw Unfiltered Monocular Gaze	±28.4 px (unusable)	0 ms	41.2% false clicks
5-Frame Moving Average Filter	±11.2 px (moderate)	165 ms (sluggish)	18.5% false clicks
Linear Kalman Filter (Constant Velocity)	±8.6 px (acceptable)	110 ms (overshoot)	11.4% false clicks
Vakyam 1€ Adaptive Filter + IMU Gate	±3.1 px (−89.1% jitter)	18 ms (< 1 frame)	1.8% false clicks

V. BEDSIDE HARDWARE SENSOR SAFEGUARD STATE MACHINES

A. Ward Illumination and Telemetry Interface Architecture

Unlike laboratory eye trackers operating under controlled infrared illumination, a smartphone clamped above a hospital bed must survive dark 2:00 A.M. wards, vibrating bed rails, nurses leaning across the camera line-of-sight, summer ambient temperatures exceeding 40 °C, and gradual patient pillow slump. Fig. 5 illustrates Vakyam's autonomous night ring-light and five-sensor diagnostic heads-up display (HUD).



(a) Autonomous Night Ring-Light (< 10 lx) (b) Five-Sensor Telemetry & Fault HUD

Fig. 5. Environmental adaptation wireframes. (a) Warm-white (`#FFFDF7`) 48-dp screen-border ring-light triggered automatically when ambient illuminance falls below 10.0 lx, illuminating the pupil without disturbing neighboring ward patients. (b) Real-time five-sensor diagnostic overlay displaying illuminance, 6-axis IMU tremor velocity, infrared proximity state, SoC thermal headroom, and 3-D nose-bridge slump offset.

B. Five Deterministic Hardware Safeguard Finite-State Machines (IEC 62304 Class B)

All five environmental watchdogs execute as nonblocking evaluators (≤ 0.8 ms combined overhead per 33-ms camera frame) synchronized via Kotlin `StateFlow` primitives:

1) *FSM 1 (Ambient Light Photodiode and CameraX Exposure Control, `Sensor.TYPE_LIGHT`):* Transition S_0 (Daylight) → S_1 (Dark Ward) occurs when ambient illuminance E_v < 10.0 lx is sustained for ≥ 1200 ms, activating the 48-dp warm-white (`#FFFDF7`) screen-border ring-light and locking CameraX exposure compensation (+2.0 EV). Hysteresis recovery $S_1 \rightarrow S_0$ requires $E_v \geq 18.0$ lx sustained for ≥ 2000 ms, strictly preventing self-illumination oscillation flicker.

2) *FSM 2 (Six-Axis IMU Bed-Tremor Freeze and Mount-Tilt Stabilizer, `TYPE_GYROSCOPE` + `ACCELEROMETER`):* Transition $S_0 \rightarrow S_1$ (Bed Shake / Cough Spasm) triggers when angular velocity $\|\boldsymbol{\omega}_{\text{gyro}}\|_2 > 0.45$ rad/s or linear acceleration $|a_{\text{lin}}| > 1.8$ m/s^2, freezing radial dwell progress at its current percentage rather than resetting patient effort to zero. Transition $S_0 \rightarrow S_2$ (Mount Sag > 12°) tracks gravity-vector pitch/roll (Δθ), geometrically translating the 6-point gaze centroid without recalibration.

3) *FSM 3 (Infrared Proximity Caregiver Occlusion Gate, `Sensor.TYPE_PROXIMITY`):* Transition $S_0 \rightarrow S_1$ (Caregiver Reach-Over) triggers when proximity distance d < 5.0 cm, immediately pausing the gaze cursor and muting TTS audio while a nurse or attendant wipes the patient's face or adjusts ventilator tubing. Recovery $S_1 \rightarrow S_0$ requires d > 5.0 cm for 500 ms plus valid 468-point facial mesh re-acquisition.

4) *FSM 4 (SoC Thermal Throttling and Unplugged Battery Watchdog, `ThermalService` + `BatteryManager`):* Transition $S_0 \rightarrow S_1$ (Thermal Warning ≥ 42.0 °C) steps down CameraX frame analysis from 30 frames/s to 15 frames/s to cool the SoC during peak summer ward temperatures. Transition $S_1 \rightarrow S_2$ (Critical ≥ 45.0 °C or Battery ≤ 15%) broadcasts a loud bilingual voice alert every 5 min to summon bedside caregiver assistance.

5) *FSM 5 (3-D Nose-Bridge Head-Slump Centroid Auto-Recentering, Landmarks #1 and #168):* Transition $S_0 \rightarrow S_1$ (Pillow Slump ≤ 18% ROI Shift) continuously translates calibration region boundaries relative to 3-D nose-bridge anchors (#1, #168) as patients slide down hospital pillows. Transition $S_1 \rightarrow S_2$ (Complete Face Loss > 8.0 s) renders a high-contrast visual alignment guide and plays an acoustic chime for caregiver repositioning.

TABLE V
HARDWARE SENSOR SAFEGUARD THRESHOLDS, HYSTERESIS WINDOWS, AND DETERMINISTIC ACTIONS

Finite-State Machine	Activation Threshold ($S_0 \rightarrow S_1$)	Recovery Hysteresis ($S_1 \rightarrow S_0$)	Deterministic Clinical Safeguard
FSM 1: Ambient Photodiode	E_v < 10.0 lx for ≥ 1200 ms	$E_v \geq 18.0$ lx for ≥ 2000 ms	48-dp `#FFFDF7` ring-light + 2.0 EV boost.
FSM 2: 6-Axis IMU Gate	$\|\boldsymbol{\omega}\|_2 > 0.45$ rad/s or $\|a\| > 1.8$ m/s^2	$\|\boldsymbol{\omega}\|_2 \leq 0.20$ rad/s for 300 ms	Freezes dwell ring progress without reset.
FSM 3: IR Proximity Gate	Distance d < 5.0 cm (immediate)	d > 5.0 cm for 500 ms + mesh	Pauses gaze cursor and mutes false TTS.
FSM 4: Thermal / Battery	$T_{\text{SoC}} \geq 42.0$ °C / Batt ≤ 15%	$T_{\text{SoC}} \leq 39.5$ °C / Charger AC	Throttles 30→15 FPS; vocalizes ward alert.
FSM 5: 3-D Head-Slump	Nose shift ≤ 18% / Loss > 8.0 s	Anchor lock ≥ 15 frames	Translates ROI; chimes on complete loss.

VI. PARTICIPATORY CLINICAL CO-DESIGN, FMEA, AND VERIFICATION MATRIX

A. Participatory Clinical Co-Design in Maharashtra (Pune Field Insights)

Vakyam was iteratively refined through direct bedside evaluation with ALS patients and neuro-rehabilitation speech-language pathologists in Pune, Maharashtra (Fig. 6). Five critical engineering modifications originated directly from bedside testing: 1) replacing 9-point calibration with a 30-s 6-point spoken countdown to prevent corner-fixation tearing; 2) freezing rather than resetting dwell progress during bed tremors; 3) tuning Gemma 4 prompts to honor Marathi and Hinglish kinship terms (*"Aai," "Kaka," "Doctor"*); 4) vocalizing low-battery and thermal alerts aloud in Marathi/Hindi; and 5) adding the 28-sp BiFocal top readout banner for ward nurses.



(a) Bedside Co-Design with ALS Patient (Pune, Maharashtra)



(b) National Jury Recognition & Validation

Fig. 6. Field evaluation and clinical recognition. (a) Bedside testing of gooseneck-mounted smartphone gaze tracking with an ALS patient in Pune. (b) National technical jury recognition validating Vakyam's offline assistive architecture.

B. Clinical Failure Mode and Effects Analysis (FMEA)

TABLE VI
CLINICAL FAILURE MODE AND EFFECTS ANALYSIS (FMEA) AND AUTOMATED MITIGATIONS

Failure Mode	Clinical Hazard	Detection Mechanism	Automated Fail-Safe Mitigation
Gemma 4 OOM on 4-GB RAM	Loss of AI sentence synthesis	`ActivityManager` RAM gate	Zero-latency fallback to offline Trie + Quick-Chat TTS.
Overhead Tube-Light Glare	Iris washout in upper basins	OpenCV MSE variance spike	CLAHE local equalization + 6-point MSE template fusion.
Nighttime Ward Power Cut	Camera loses facial mesh lock	`TYPE_LIGHT` < 10.0 lx	Auto-enables 48-dp `#FFFDF7` screen-border ring-light.
Unilateral Facial Palsy	Affected eye remains closed	EAR asymmetry detector	Binds gaze and blink pipeline exclusively to healthy eye.
Caregiver Suction Reach-Over	Erratic false speech triggers	`TYPE_PROXIMITY` < 5.0 cm	Freezes cursor and mutes TTS until line-of-sight clears.
Phone Overheats (≥ 45.0 °C)	Thermal shutdown on pillow	`ThermalService` watchdog	Throttles camera to 15 FPS + speaks caregiver voice alert.
Ptosis / Eyelid Droop Fatigue	False long-blink undo actions	Rolling 60-s EAR_0 tracker	Re-anchors open-eye baseline EAR_0 per individual eye.
ICU Ventilator Acoustic Hum	ASR misinterprets pump noise	Spectral bandpass gate	300 Hz–3.4 kHz voice filter attenuates machine hiss.

C. Automated Quality Assurance and Repository Verification (294 JVM Unit Tests)

The Vakyam repository is verified by 294 deterministic JVM unit tests executing on GitHub Actions (JDK 17, Robolectric 4.11 headless Android) with 94.2% line coverage, 100% safety FSM branch coverage, and zero failures:

TABLE VII
AUTOMATED JVM UNIT TEST SUITE BREAKDOWN AND VERIFICATION COVERAGE

Verification Domain	Test Count	Core Test Suites and Verified Engineering Invariants
1. Vision & Signal Filtering	78 tests	`OneEuroFilterTest` (16), `GazeEnginePipelineTest` (24), `BlinkDetectorTest` (22), `CalibrationValidatorTest` (16): verifies step lag, MSE+Iris fusion, blink bounds, NaN/Inf resilience.
2. Typing & Edge-AI Loop	112 tests	`FourGroupTriePredictorTest` (38), `UserHistoryBoostTest` (26), `GemmaSentenceGeneratorTest` (28), `WhisperTranscriberTest` (20): verifies $O(L)$ pruning, 14-day decay, INT4 fallback.
3. Hardware Safeguard FSMs	104 tests	`AmbientLightRingTest` (22), `ImuTremorGateTest` (28), `ProximityGuardTest` (18), `ThermalBatteryTest` (36): verifies 10-lx/18-lx hysteresis, 0.45-rad/s freeze, < 5-cm mute, 42 °C throttle.

Clinical Ethics and Data Governance: All facial video frames and Whisper audio buffers are processed strictly in volatile RAM and discarded within 33 ms. Zero camera frames, voice prints, or transcripts ever leave the handset.